

DATA QUALITY ASSESSMENT

RAW DATA QUALITY AUDIT

- i. Missing Values: 19 missing values (3.83×10^{-6} missing rate) are present in the dataset, all in the 10degC/582_LA92.csv file (90740 out of 90740 rows). The NaN values are located in the file at rows 1491, 1492, 1943, and 1944 (in Status = TABLE mode), not during t=0 or channel initialization.
- ii. Duplicate Rows: Exactly 5,876 duplicate rows (0.119% of total) are present in the data set. These are found primarily during transitions between cyler pauses (Status = PAU) where the measurement sensors record identical values at consecutive ticks.
- iii. Timestamp Monotonicity and Dates Analysis: In the majority of files, timestamps appear to follow a strictly increasing order. It appears that only 582_LA92.csv has 4 entries (indices 229, 1488, 1491, and 1943) that have timestamps that are not strictly greater than the previous entry. Across all file headers, the timestamp ranges from October 25, 2018, at 4:09:59 AM to December 26, 2018, at 10:37:50 PM.
- iv. Invalid Ranges and Voltage Anomalies There are three files in which one sample of the transient had dropped to 0.000V (0degC/590_Charge16.csv, 25degC/552_Charge16.csv, and 40degC/557_Charge10.csv). Excluding these three outliers, the terminal voltage ranges from 2.000V (discharge cutoff) to 4.2399V (charge cutoff). The discharge current reaches -18.098A (~6C rate), while the charge current reaches +6.005A (2C rate).
- v. Thermal Chamber Ranges: Temperature range -22.50 °C (in -20 °C regime) to +41.33 (in 40 °C regime). For the 25 °C base folder the cell surface temperature reaches a maximum of 27.87 °C (never goes above 27.87).
- vi. Categorical status codes and procedure schema: The raw Status column has exactly five unique codes: CHA (Charge), DCH (Discharge coded as DCH, not DIS), PAU (Pause/Rest), STO (Stop), and TABLE (Dynamic Drive Cycle). The row-level Procedure column has six generic cyler script names (HPPC_4pulse, HPPC_Multi_Pulse, LG_HG2_CyclesA,

LG_HG2_CyclesB, LG_HG2_NN_Char, NN_Char_Charge). The specific drive cycle name (UDDS, LA92, US06, HWFET) resides in the file header metadata field (Test section); not a row CSV column.

- vii. Sampling Rates: Dynamic profile steps (Status = TABLE) are sampled at uniform 0.1-s intervals (10 Hz). Constant current/voltage charging profiles (Charge.csv) are sampled on ~60-s intervals (median Δt = 60.0s, mean Δt = 59.1s-59.9s).

Data Quality Table

Feature Name	Data Type & Unit	Missing Count	Expected Range	Observed Range (Min / Max)	Duplicate / Anomaly	Quality Status & Leakage Risk
Time Stamp	Datetime (String)	0 (0.00%)	Monotonic wall-clock	10/25/2018 to 12/26/2018	4 non-mono jumps in LA92	Clean / Base Index. Needs sorting in 582_LA92.
Step	Int64	0 (0.00%)	$1 \leq \text{Step} \leq 9999$	1.0 to 9999.0	Sequential step index	Moderate Risk. Cyclor step marker.
Status	Categorical	0 (0.00%)	5 codes: CHA, DCH, PAU, STO, TABLE	CHA, DCH, PAU, STO, TABLE	5,876 dup rows at PAU	Clean. Operational phase marker (DCH = Discharge).
Prog Time	String / Time	0 (0.00%)	Monotonic prog time	00:00:00 to 24:00:00+	Resets at program start	Clean. Used for time step Δt calculation.
Step Time	String / Time	0 (0.00%)	Monotonic step time	00:00:00 to 02:30:00	Resets per step change	Moderate Risk. Resets on step transition.
Cycle	Int64	0 (0.00%)	$0 \leq \text{Cycle} \leq 10$	0.0 to 10.0	Integer cycle counter	Low Risk. Meta counter.
Cycle Level	Int64	0 (0.00%)	0 or 1	0.0 to 1.0	Hierarchy level	Low Risk. Meta hierarchy.
Procedure	Categorical	0 (0.00%)	6 script names	6 script names (LG_HG2_CyclesA, etc.)	Shared across files	Low Risk in CSV. Generic script name.
Voltage	Float64 [V]	3 (6.05×10 ⁻⁷ %)	2.000V to 4.250V	0.000V to 4.2399V	3 single-sample 0V drops	Core Input Feature. Requires 0V

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						drop filtering.
Current	Float64 [A]	3 ($6.05 \times 10^{-7}\%$)	-18.10A to +6.00A	-18.0983A to +6.0047A	Smooth dynamic profiles	Core Input Feature. +I = Charge, -I = Discharge.
Temperature	Float64 [°C]	3 ($6.05 \times 10^{-7}\%$)	-20.0°C to +45.0°C	-22.5039°C to +41.3273°C	Smooth surface temp	Core Input Feature. Max 27.87°C in 25°C folder.
Capacity	Float64 [Ah]	3 ($6.05 \times 10^{-7}\%$)	-3.00 Ah to +3.00 Ah	-2.7807 Ah to +2.8180 Ah	Cumulative cyler Ah	CRITICAL LEAKAGE. Directly leaks target Ah!
WhAccu	Float64 [Wh]	3 ($6.05 \times 10^{-7}\%$)	-12.0 Wh to +12.0 Wh	-10.3028 Wh to +10.9525 Wh	Cumulative cyler Wh	CRITICAL LEAKAGE. Leaks target energy Wh!
Cnt	Float64	4 ($8.07 \times 10^{-7}\%$)	$1 \leq \text{Cnt} \leq 51$	1.0 to 51.0	Internal sample counter	Low Risk / Irrelevant. Drop during modeling.
SoC (%) (Target)	Float64 [%]	Derived	0.0% to 100.0%	0.0% to 100.0%	Coulomb counted target	Primary Target Variable. Integrated current.

Condition specific problem and Daat leakage identification

- Pause Step Duplicate Accumulation (Status = PAU): The majority of 5,876 duplicated rows occur during extended pause steps (PAU) where voltage, current (0.0A), and temperature are identical across consecutive ticks.
- 582_LA92.csv File-Specific Anomalies: Contains 4 NaN entries at indices 1491, 1492, 1943, 1944 (during Status = TABLE execution) and 4 non-monotonic timestamp entries where tick out of order
- Regime-Dependent Operational Patterns (Sampling and Thermal): Dynamic drive cycle profiles (Status=TABLE) sampled at uniform 0.1s intervals (10Hz) while charging steps (Charge.csv) sampled every ~60s (median $\Delta t = 60.0s$). Thermal similarly tracks ambient regime: temp reaches 27.87C ambient 25C, 41.33C ambient 40C and -22.50C ambient -20C.

- Target Ah/Wh Accumulation Leakage (Capacity & WhAccu): Capacity and WhAccu record cumulative integrated Ah and Wh from step start. Providing these as model inputs directly leaks the integrated target value (SoC), creating artificial near-perfect training scores ($R^2 > 0.999$) that collapse in real BMS deployment. Remedy: Drop these entirely from model input feature matrices.
- Drive Cycle Metadata vs. Procedure Script Name: The row-level Procedure column stores generic script names (LG_HG2_CyclesA, HPPC_4pulse, etc.) that do NOT leak the specific drive cycle identity. The actual drive cycle name (UDDS, LA92, US06, HWFET) is isolated in the file header metadata field (Test). It must be used strictly for grouping/splitting validation folds, never fed as an input feature.

Proposed Cleaning rules & Technical justification

1. Duplicate row deduplication: Remove the 5,876 exact duplicate rows occurring during PAU steps (`df.drop_duplicates()`). Duplicate rows during rest steps over-represent zero-current equilibrium states without adding new dynamic information.
2. Mid-File missing value Imputation: Apply linear interpolation (`df.interpolate(method='linear')`) for the 19 NaN entries at indices 1491-1944 in 582_LA92.csv. Voltage, current, and temperature are continuous physical quantities. Mid-file linear interpolation restores smooth time-series continuity.
3. Timestamp & Sorting: Sort rows by Time Stamp in 582_LA92.csv to fix the 4 out-of-order timestamp entries, then re-compute Δt . Out-of-order timestamps break trapezoidal integration for Coulomb counting and time-derivatives of voltage and current (dV/dt , dI/dt).
4. Replace single-sample 0.000V drops in 0degC/590_Charge16, 25degC/552_Charge16, and 40degC/557_Charge10 with linear interpolation bounded by [2.000V, 4.250V]. Terminal voltage cannot drop instantaneously to 0.000V and recover to >4.0V in 0.1s. These are hardware sensor acquisition dropouts.
5. Uniform Time Grid Resampling: Resample dynamic profiles to a uniform 0.1s time grid ($\Delta t = 0.1s$) and charging profiles to 1.0s or 60.0s uniform grid using linear interpolation for V, I, T and zero-order hold for Status.

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Standardizes sampling intervals across dynamic and charging profiles for machine learning architectures (LSTM, CNN-LSTM).

6. Target accumulation exclusion: Exclude Capacity and WhAccu from input feature space. Use instantaneous $V(t)$, $I(t)$, $T(t)$ and causal lagged features (V_{t-k} , I_{t-k} , T_{t-k} , dV/dt , dI/dt , dT/dt). Prevents target leakage from cumulative cyclers Ah/Wh integrals, ensuring model generalization to real-world embedded BMS applications.